Data Science Glossary

Data prep and pitfalls

Train/test split: hold out ~20% to score on unseen data.

Stratify: keep the class mix the same in train and test.

random_state: fixed seed so splits and results repeat.

Standardization: rescale to mean 0, SD 1; coefficients become per-SD.

MinMax / Robust scaling: squeeze to 0-1 / scale by median and IQR (outlier-resistant).

Centering: subtract the mean (cut production VIF 4.49 to 3.37).

Log transform: log(1+x) tames right skew (accident cost).

Dummy variable: 0/1 column per category (ownership type).

Lag feature: yesterday's value predicting today (futures).

Data leakage: answer information sneaks into the predictors (production gave F1 0.998).

Circular target: target is arithmetic from a predictor (emissions = 0.3714 x production).

Multicollinearity: predictors move together.

VIF: variance inflation from overlap; above ~5-10 is a concern (crude 383-806; sensors ~1).

Pipeline: scikit-learn wrapper keeping scaling inside each CV fold.

Regression and fit

Coefficient: change in prediction per unit of a feature, others fixed.

Residual: actual minus predicted.

R^2: share of target variance explained (0 = no better than the mean, 1 = perfect).

Adjusted R^2: R^2 penalized for extra predictors.

RMSE: typical error, in the target's units.

Std. error / t-stat: coefficient's uncertainty / estimate divided by it.

p-value: chance of a result this strong if no true effect (<0.05 = unlikely chance).

ANOVA / F-test: do added terms improve fit more than chance?

Polynomial term: squared/cubed feature that curves the fit.

Interaction: two features multiplied; one's effect depends on the other.

In- vs out-of-sample: scored on the data used to fit vs on unseen data.

Overfitting and regularization

Overfitting: memorizing noise: strong train, weak test.

Underfitting: too simple for the real pattern.

Generalization: doing well on new data (the goal).

Bias-variance tradeoff: simple = stable but misses patterns (bias); flexible = fits patterns but chases noise (variance).

Signal vs noise: tuning cannot create signal that isn't there.

Regularization: penalty on large coefficients.

Alpha: penalty strength (0 = ordinary regression; larger = more shrinkage).

Ridge (L2): shrinks coefficients, keeps all features.

Lasso (L1): can zero out features.

Elastic net / l1_ratio: blend (1.0 = lasso, 0.0 = ridge).

Selection and dimension reduction

Forward/backward selection: add/remove features one at a time by model improvement.

PCA: combines correlated features into uncorrelated components ordered by variance.

Explained variance ratio: share of total variance a component holds (2 PCs = 77%).

PCR: regression on components built without looking at the target.

PLSR: components built to explain the target.

Classification and metrics

Logistic regression: linear score (log-odds) through a sigmoid gives P(class 1).

Decision threshold: cutoff (usually 0.5) turning probability into yes/no.

Median split: above/below median gives balanced classes, so accuracy is fair.

Baseline / chance: the bar to beat (accuracy ~0.5, AUC 0.5).

Confusion matrix: counts of true/false positives and negatives.

Accuracy: share correct.

Precision: of predicted positives, share truly positive.

Recall: of true positives, share found.

F1: harmonic mean of precision and recall; inflated if a model predicts 'positive' almost always.

ROC-AUC: chance a random positive ranks above a random negative; 0.5 = coin flip, 1.0 = perfect; threshold-free.

C: regularization for logistic/SVM; small C = stronger (opposite of alpha).

k-nearest neighbors (KNN): classifies a point by majority vote of its k closest training points (regression: their average). Distance-based, so scale features first. Small k = flexible, overfits; large k = smoother. k is a hyperparameter; there is no training step.

Support vector machines

Hyperplane / margin: the boundary and the gap to the nearest points; SVM maximizes the gap.

Support vectors: points nearest the boundary; only they set it.

Kernel trick: curved boundaries without computing new features.

Linear vs RBF: straight boundary vs flexible curve.

Gamma: reach of one point's influence in RBF (high = wiggly).

Trees and forests

Decision tree: chain of yes/no splits ending in a prediction.

Gini impurity: 1 - sum(p^2); 0 = pure, 0.5 = 50/50 mix (default split rule).

Entropy: -sum(p log2 p); 0 = pure, 1 = 50/50; usually gives similar trees.

Log loss: -mean[y ln p + (1-y) ln(1-p)]; punishes confident wrong probabilities. As a tree split rule (criterion='log_loss' in scikit-learn) it is the same as entropy.

max_depth: levels of splits allowed (deeper = more overfit).

min_samples_leaf: minimum points per final group (larger = smoother).

Pruning: limiting/trimming a tree (depth, leaf size) to curb overfitting.

Bootstrap / bagging: train each tree on a random resample of rows, then average.

Random forest: many bagged trees with random feature subsets, votes averaged.

n_estimators: number of trees.

max_features (sqrt): features considered per split.

Feature importance: total impurity reduction per feature; favors continuous features.

Validation and tuning

Cross-validation: split training data into k folds; train on k-1, test on the held-out fold, rotate.

Hyperparameter: setting chosen before training (alpha, C, depth).

GridSearchCV: tries every combination with CV and keeps the best.

Validation curve: error plotted against one hyperparameter.

Train/test gap: train score minus test score; large = overfit.

Efficient-market idea: prices already reflect past information, so lagged prices shouldn't predict direction.